

SHREEDHAR JOSHI

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GitHub | LinkedIn | Codeforces Profile

EDUCATION

Indian Institute of Information Technology (IIIT), Surat

2025 – Expected 2029

B.Tech - Electronics and Communication Engineering

GPA (2nd Sem): 8.05/10.0

- **Core Courses:** Data Structures & Algorithms, C Programming, Discrete Mathematics.

Indian Institute of Technology (IIT), Madras

Jul 2025 – Present

BS - Data Science and Applications

GPA: 7.88/10.0

- **Core Focus:** Computational Thinking, Mathematical Foundations, Statistics, Machine Learning, Business Data Mgt.

TECHNICAL SKILLS

- **Languages:** C (Primary), Python, Java, TypeScript, JavaScript, Embedded C, HTML/CSS, SQL (PostgreSQL).
- **Frameworks & Libraries:** React.js, Tailwind CSS v4, @dnd-kit, Node.js, Express.js, MongoDB (Mongoose), NumPy.
- **Developer Tools & Software:** Docker, ClamAV, DBMS, Android Studio, Proteus, Git/GitHub, Arduino IDE, Atmel AVR, WebAssembly (WASM), JWT.
- **Platforms & Deployment:** Supabase (BaaS), Render, GitHub Pages, Netlify.
- **Hardware & Prototyping:** Arduino Microcontrollers, Sensor Interfacing, Basic Circuit Design.
- **Interests:** Competitive Programming, Astronomy, Robotics, Digital Audio Production.

EXPERIENCE & LEADERSHIP

Atmiya University (e-Yantra, IIT Bombay Initiative) | Summer Intern – Robotics & Embedded Systems Jun 2026 – Jun 2026

- Programmed microcontroller firmware utilizing **Embedded C** and worked with bare-metal registers on **Atmel AVR** chips.
- Designed and successfully simulated complex circuits and peripheral interfaces using **Proteus**.
- Collaborated at the Center of Excellence to interface analog/digital sensors and process live hardware feedback loops.

India Space Lab | Summer Intern

Jun 2026 – Jul 2026

- Trained in advanced drone technologies, CanSat design, and CubeSat satellite architectures.
- Developed embedded control algorithms for simulated payload flight systems.

ASTRA (Official Space Sciences Club, IIIT Surat) | Founder & Student Co-ordinator

Apr 2026 – Present

- Established the first astrophysics community at IIIT Surat; authored the organization's charter and framework.
- Spearheaded recruitment and successfully executed the orientation for incoming engineering student members.

TECHNICAL PROJECTS

SafeGate (Browser-Assisted Cloud Sandbox) | Python, Docker, ClamAV, PostgreSQL

June 2026

- Engineered a secure virtualization utility designed to safely analyze and inspect suspicious binaries and file downloads.
- Built isolated containment environments using **Docker** to spin up dynamic, secure runtimes for analysis verification.

Northstar Kanban Board (Real-Time Task Management) | React, TypeScript, Supabase, Tailwind v4

June 2026

- Engineered a collaborative agile workspace utilizing **Supabase Realtime Channels** to achieve (<50ms) state synchronization.
- Designed an asynchronous **6-second undo transaction buffer** for optimistic layout UI state mutations, decoupling user interactions to reduce accidental data deletion.
- Optimized rendering bottlenecks by applying memoization (React.memo) and callback caching.

Ultrasonic Length Measurement System | Embedded C, Arduino, Sensors, Proteus

Jul 2026

- Engineered an automated scanning mechanism using Arduino and a servo motor sweeping between 20° and 160° to map environments.
- Implemented dynamic edge detection and distance-averaging algorithms in Embedded C to filter sensor noise and identify boundaries.
- Applied coordinate conversions and geometric formulas to estimate physical dimensions.

HONORS, AWARDS & CERTIFICATIONS

- **State Topper | Indian Olympiad Qualifier in Junior Science (NSEJS):** Achieved a Top 1% State Rank in the National Standard Examination; recognized for proficiency in Physics, Chemistry, and Biology (May 2022).
- **INSPIRE Awardee (DST) - MANAK:** Recipient of DST national award and a grant of INR 10,000 for innovative science pursuit (Jan 2023).
- **Gemini Certified University Student (Google for Education):** Professional credential in Prompt Engineering and GenAI (Jul 2026).